

《图解大模型：生成式 AI 原理与实战》

链接资源¹

前言

- <https://www.learnpython.org/>
LearnPython 网站
- <https://colab.research.google.com/>
Google Colab 官网
- <https://jalammar.github.io/illustrated-transformer/>
Jay Alammar 的文章 “Illustrated Transformer”
- <https://platform.openai.com/>
OpenAI 开发者平台
- <https://dashboard.cohere.com/>
Cohere 开发者平台
- <https://huggingface.co/>
Hugging Face 官网

1.1 节

- <http://jmc.stanford.edu/artificial-intelligence/what-is-ai/index.html>
<https://cse.unl.edu/~choueiry/S09-476-876/Documents/whatisai.pdf>
John McCarthy 的文章 “What is Artificial Intelligence?”

1.2 节

- <https://arxiv.org/abs/1409.0473>
Dzmitry Bahdanau, Kyunghyun Cho, and Yoshua Bengio. “Neural Machine Translation by Jointly Learning to Align and Translate.” arXiv preprint arXiv:1409.0473 (2014).
- <https://arxiv.org/abs/1706.03762>
Ashish Vaswani et al. “Attention is All You Need.” Advances in Neural Information Processing Systems 30 (2017).

注 1：由于项目更新迭代，原书所提供链接内容可能过时或失效，仅供参考。——编者注

- https://s3-us-west-2.amazonaws.com/openai-assets/research-covers/language-unsupervised/language_understanding_paper.pdf
Alec Radford et al. “Improving Language Understanding by Generative Pre-training”, (2018)
【截至本书出版，链接已失效】
- <https://newsletter.maartengrootendorst.com/p/a-visual-guide-to-mamba-and-state>
“A Visual Guide to Mamba and State Space Models”

1.6 节

- <https://artificialintelligenceact.eu/>
欧盟《人工智能法案》官方页面

1.8 节

- <https://github.com/openai/openai-python>
openai-python 的 Github 仓库
- <https://github.com/ggml-org/llama.cpp>
llama.cpp 的 Github 仓库
- <https://github.com/langchain-ai/langchain>
LangChain 的 Github 仓库
- <https://github.com/huggingface/transformers>
Transformers 的 Github 仓库
- <https://github.com/oobabooga/text-generation-webui>
text-generation-webui 的 Github 仓库
- <https://github.com/LostRuins/koboldcpp>
KoboldCpp 的 Github 仓库
- <https://lmstudio.ai/>
LM Studio 官网

2.1 节

- <https://platform.openai.com/tokenizer>
OpenAI 分词器页面
- <https://arxiv.org/abs/2103.06874>
论文 “CANINE: Pre-training an Efficient Tokenization-Free Encoder for Language Representation”

- <https://arxiv.org/abs/2105.13626>
论文 “ByT5: Towards a Token-Free Future with Pre-trained Byte-to-Byte Models”
- <https://www.oreilly.com/library/view/designing-large-language/9781098150495/>
图书 *Designing Large Language Model Applications* 的 O'Reilly 官方页面
- <https://huggingface.co/google-bert/bert-base-uncased>
BERT 基座模型（大小写不敏感）的 Hugging Face 模型仓库页面
- <https://ieeexplore.ieee.org/document/6289079>
论文 “Japanese and Korean Voice Search”
- <https://huggingface.co/google-bert/bert-base-cased>
BERT 基座模型（大小写敏感）的 Hugging Face 模型仓库页面
- <https://huggingface.co/openai-community/gpt2>
GPT-2 的 Hugging Face 模型仓库页面
- <https://arxiv.org/abs/1508.07909>
论文 “Neural Machine Translation of Rare Words with Subword Units”
- <https://huggingface.co/google/flan-t5-xxl>
Flan-T5 XXL 的 Hugging Face 模型仓库页面
- <https://arxiv.org/pdf/1808.06226>
论文 “SentencePiece: A Simple and Language Independent Subword Tokenizer And Detokenizer for Neural Text Processing”
- <https://arxiv.org/abs/1804.10959>
论文 “Subword Regularization: Improving Neural Network Translation Models with Multiple Subword Candidates”
- <https://arxiv.org/abs/2207.14255>
论文 “Efficient Training of Language Models to Fill In the Middle”
- <https://huggingface.co/bigcode/starcoder2-15b>
StarCoder2-15B 的 Hugging Face 模型仓库页面
- <https://arxiv.org/abs/2402.19173>
论文 “StarCoder 2 and The Stack v2: The Next Generation”
- <https://arxiv.org/abs/2305.06161>
论文 “StarCoder: May the Source be with You!”
- <https://huggingface.co/facebook/galactica-1.3b>
GALACTICA 1.3B 的 Hugging Face 模型仓库页面

- <https://arxiv.org/abs/2211.09085>
论文 “GALACTICA: A Large Language Model for Science”
- <https://huggingface.co/microsoft/Phi-3-mini-4k-instruct>
Phi-3-Mini-4K-Instruct 的 Hugging Face 模型仓库页面
- <https://huggingface.co/meta-llama/Llama-2-7b-hf>
Llama 2 7B 的 Hugging Face 模型仓库页面
- https://huggingface.co/docs/transformers/tokenizer_summary
Hugging Face 的分词器汇总页面
- <https://huggingface.co/learn/nlp-course/chapter6/1?fw=pt>
Hugging Face 上的 NLP 课程中分词器相关部分
- <https://www.oreilly.com/library/view/natural-language-processing/9781098136789/>
图书 *Natural Language Processing with Transformers, Revised Edition* 的 O'Reilly 官方页面

2.2 节

- <https://www.bbc.com/news/technology-64538604>
关于 “Google 杀手” 的文章
- <https://openreview.net/forum?id=sE7-XhLxHA>
论 文 “DeBERTaV3: Improving DeBERTa Using ELECTRA-Style Pre-training with Gradient-Disentangled Embedding Sharing”

2.3 节

- <https://github.com/UKPLab/sentence-transformers>
sentence-transformers 的 Github 仓库
- <https://huggingface.co/sentence-transformers/all-mpnet-base-v2>
all-mpnet-base-v2 模的 Hugging Face 模型仓库页面

2.4 节

- <https://radimrehurek.com/gensim/>
Gensim 官网
- <https://arxiv.org/abs/1301.3781>
论文 “Efficient Estimation of Word Representations in Vector Space”
- <https://jalammar.github.io/illustrated-word2vec/>
文章 “The illustrated word2vec”

- <https://proceedings.mlr.press/v9/gutmann10a/gutmann10a.pdf>
论文 “Noise-Contrastive Estimation: A New Estimation Principle for Unnormalized Statistical Models”

2.5 节

- https://www.cs.cornell.edu/~shuochen/lme/data_page.html
Playlist 数据集

3.1 节

- <https://kipp.ly/transformer-inference-arithmetic/>
文章 “Transformer Inference Arithmetic”，其中介绍了键 - 值缓存相关内容
- <https://web.stanford.edu/~jurafsky/slp3/>
Speech and Language Processing (3rd ed. draft) 在线阅读

3.2 节

- <https://arxiv.org/pdf/1904.10509>
论文 “Generating Long Sequences with Sparse Transformers”
- <https://arxiv.org/pdf/2004.05150>
论文 “Longformer: The Long-Document Transformer”
- <https://arxiv.org/abs/2305.13245>
论文 “GQA: Training Generalized Multi-Query Transformer Models from Multi-Head Checkpoints”
- <https://arxiv.org/abs/1911.02150>
论文 “Fast Transformer Decoding: One Write-Head is All You Need”
- <https://arxiv.org/abs/2205.14135>
论文 “FlashAttention: Fast and Memory-Efficient Exact Attention with IO-Awareness”
- <https://tridao.me/publications/flash2/flash2.pdf>
论文 “FlashAttention-2: Faster Attention with Better Parallelism and Work Partitioning”
- <https://arxiv.org/abs/2002.04745>
论文 “On Layer Normalization in the Transformer Architecture”
- <https://arxiv.org/abs/1910.07467>
论文 “Root Mean Square Layer Normalization”
- <https://arxiv.org/pdf/2002.05202>
论文 “GLU Variants Improve Transformer”

- <https://arxiv.org/abs/2104.09864v4>
论文 “RoFormer: Enhanced Transformer with Rotary Position Embedding”
- <https://arxiv.org/abs/2107.02027>
论文 “Efficient Sequence Packing without Cross-Contamination: Accelerating Large Language Models without Impacting Performance”
- <https://www.graphcore.ai/posts/introducing-packed-bert-for-2x-faster-training-in-natural-language-processing>
视频 + 文章 “Introducing Packed BERT for 2X Training Speed-Up in Natural Language Processing”
- <https://arxiv.org/abs/2106.04554>
论文 “A Survey of Transformers”
- <https://dl.acm.org/doi/abs/10.1145/3505244>
论文 “Transformers in Vision: A Survey”
- <https://ieeexplore.ieee.org/abstract/document/9716741>
论文 “A Survey on Vision Transformer”
- <https://robotics-transformer-x.github.io/>
论文 “Open X-Embodiment: Robotic Learning Datasets and RT-X Models”
- <https://arxiv.org/abs/2202.07125>
论文 “Transformers in Time Series: A Survey”

4.1 节

- <https://huggingface.co/datasets>
Hugging Face 的数据集仓库主页
- https://huggingface.co/datasets/cornell-movie-review-data/rotten_tomatoes
rotten_tomatoes 数据集的 Hugging Face 页面

4.3 节

- https://huggingface.co/models?pipeline_tag=text-classification
Hugging Face 上用于文本分类的模型一览
- https://huggingface.co/models?pipeline_tag=feature-extraction
Hugging Face 上生成嵌入向量（特征提取）的模型一览
- <https://huggingface.co/FacebookAI/roberta-base>
RoBERTa 基座模型的 Hugging Face 模型仓库页面

- <https://huggingface.co/distilbert/distilbert-base-uncased>
DistilBERT 基座模型（大小写不敏感）的 Hugging Face 模型仓库页面
- <https://huggingface.co/microsoft/deberta-base>
DeBERTa 基座模型的 Hugging Face 模型仓库页面
- <https://huggingface.co/prajjwal1/bert-tiny>
bert-tiny 的 Hugging Face 模型仓库页面
- <https://huggingface.co/albert/albert-base-v2>
ALBERT base v2 的 Hugging Face 模型仓库页面
- <https://huggingface.co/cardiffnlp/twitter-roberta-base-sentiment-latest>
Twitter-roBERTa-base for Sentiment Analysis 的 Hugging Face 模型仓库页面
- <https://huggingface.co/spaces/mteb/leaderboard>
MTEB 排行榜

4.4 节

- <https://huggingface.co/distilbert/distilbert-base-uncased-finetuned-sst-2-english>
基于 DistilBERT base 大小写不敏感版本的微调模型 SST-2（DistilBERT base uncased finetuned SST-2）的 Hugging Face 模型仓库页面

4.5 节

- <https://huggingface.co/tasks/zero-shot-classification>
关于零样本分类的介绍

4.6 节

- <https://arxiv.org/abs/2210.11416>
论文 “Scaling Instruction-Finetuned Language Models”
- <https://openai.com/index/chatgpt/>
OpenAI 对 ChatGPT 的介绍，包含对其训练过程的概述
- <https://openai.com/>
OpenAI 官网
- <https://platform.openai.com/api-keys>
OpenAI API 密钥管理页面
- <https://platform.openai.com/docs/guides/rate-limits/retrying-with-exponential-backoff>
OpenAI 关于使用指数退避策略进行请求重试的指南

5.1 节

- <https://arxiv.org/>
ArXiv 平台官网
- https://huggingface.co/datasets/MaartenGr/arxiv_nlp
arxiv_nlp 数据集
- <https://arxiv.org/list/cs.CL/recent>
ArXiv cs.CL（计算与语言）板块
- <https://huggingface.co/thenlper/gte-small>
gte-small 的 Hugging Face 模型仓库页面

5.3 节

- <https://maartengr.github.io/BERTopic/>
BERTopic 官方文档页面
- <https://github.com/MaartenGr/BERTopic>
BERTopic 的 Github 仓库页面
- https://maartengr.github.io/BERTopic/getting_started/best_practices/best_practices.html
BERTopic 的最佳实践指南
- <https://github.com/MaartenGr/KeyBERT>
KeyBERT 的 Github 仓库页面
- https://maartengr.github.io/BERTopic/getting_started/multiaspect/multiaspect.html
BERTopic 官方文档中关于生成不同表示的内容
- <https://github.com/TuttleInstitute/datamapplot>
datamapplot 的 Github 仓库页面

6.4 节

- <https://github.com/dave1010/tree-of-thought-prompting>
文章 “Using Tree-of-Thought Prompting to Boost ChatGPT’s Reasoning”

6.5 节

- <https://github.com/guidance-ai/guidance>
Guidance 的 Github 仓库页面
- <https://github.com/guardrails-ai/guardrails>
Guardrails 的 Github 仓库页面

- <https://github.com/eth-sri/lmql>
LMQL 的 Github 仓库页面
- <https://github.com/abetlen/llama-cpp-python>
llama-cpp-python 的 Github 仓库页面
- <https://huggingface.co/microsoft/Phi-3-mini-4k-instruct-gguf>
Phi-3-mini-4k-instruct-gguf 的 Hugging Face 模型仓库页面

第 7 章

- <https://github.com/stanfordnlp/dspy>
DSPy 的 Github 仓库页面
- <https://github.com/deepset-ai/haystack>
Haystack 的 Github 仓库页面

7.1 节

- <https://newsletter.maartengrootendorst.com/p/a-visual-guide-to-quantization>
Maarten Grootendorst 的文章 “A Visual Guide to Quantization”
- https://huggingface.co/spaces/open-llm-leaderboard/open_llm_leaderboard
Open LLM Leaderboard (开源大语言模型排行榜)
- <https://lmarena.ai/>
Chatbot Arena 官网

7.4 节

- <https://duckduckgo.com/>
DuckDuckGo 搜索引擎

第 8 章

- <https://arxiv.org/abs/1810.04805>
论文 “BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding”
- <https://blog.google/products/search/search-language-understanding-bert/>
谷歌的文章 “Understanding Searches Better than Ever Before”，其中提到 BERT 是“搜索史上最具有突破性的进步之一”
- <https://azure.microsoft.com/en-us/blog/bing-delivers-its-largest-improvement-in-search-experience-using-azure-gpus/>
微软说明 BERT 为必应搜索带来显著用户体验提升的文章

8.2 节

- [https://en.wikipedia.org/wiki/Interstellar_\(film\)](https://en.wikipedia.org/wiki/Interstellar_(film))
电影《星际穿越》英文维基百科页面
- <https://docs.cohere.com/reference/rerank>
Cohere Rerank 端点文档
- <https://cohere.com/blog/rerank-3>
Cohere 关于 Rerank 3 的介绍文章
- <https://www.sbert.net/>
Sentence Transformers 官方文档
- https://www.sbert.net/examples/applications/retrieve_rerank/README.html
Sentence Transformers 官方文档 “Retrieve & Re-Rank” 一章
- <https://arxiv.org/abs/1910.14424>
论文 “Multi-Stage Document Ranking with BERT”
- <https://arxiv.org/abs/2010.06467>
论文 “Pretrained Transformers for Text Ranking: BERT and Beyond”
- <https://nlp.stanford.edu/IR-book/html/htmledition/irbook.html>
“Introduction to Information Retrieval”
- <https://nlp.stanford.edu/IR-book/html/htmledition/evaluation-in-information-retrieval-1.html>
Introduction to Information Retrieval 一书中 “Evaluation in Information Retrieval” 一章

8.3 节

- <https://proceedings.neurips.cc/paper/2020/file/6b493230205f780e1bc26945df7481e5-Paper.pdf>
论文 “Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks”
- <https://www.perplexity.ai/>
Perplexity
- <https://copilot.microsoft.com/>
Microsoft Copilot
- <https://gemini.google.com/>
Google Gemini
- <https://huggingface.co/BAAI/bge-small-en-v1.5>
bge-small-en-v1.5 的 Hugging Face 模型仓库页面

- <https://cohere.com/blog/command-r-plus-microsoft-azure>
Cohere 介绍 Command R+ 的文章
- <https://huggingface.co/CohereForAI/c4ai-command-r-plus>
c4ai-command-r-plus (Command R+ 的开放权重版本) 的 Hugging Face 模型仓库页面
- <https://arxiv.org/abs/2304.09848>
论文 “Evaluating Verifiability in Generative Search Engines”
- <https://github.com/explodinggradients/ragas>
Ragas 的 Github 仓库页面
- <https://docs.ragas.io/en/stable/>
Ragas 官方文档

9.2 节

- https://github.com/mlfoundations/open_clip
OpenCLIP 的 Github 仓库页面

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- <https://github.com/haotian-liu/LLaVA>
LLaVA 的 Github 仓库页面
- <https://huggingface.co/mistralai/Mistral-7B-v0.1>
Mistral-7B-v0.1 的 Hugging Face 模型仓库页面
- <https://huggingface.co/HuggingFaceM4/idefics2-8b>
Idefics2-8b 的 Hugging Face 模型仓库页面

10.4 节

- <https://gluebenchmark.com/>
GLUE 基准测试集
- <https://www.sbert.net/docs/training/overview.html#best-transformer-model>
SBERT 关于最佳 Transformer 模型的说明文档
- https://www.sbert.net/docs/package_reference/sentence_transformer/losses.html
SBERT 损失函数文档

10.5 节

- https://www.sbert.net/docs/sentence_transformer/pretrained_models.html
sentence-transformers 库的文档页面，关于预训练模型的部分

10.6 节

- <https://arxiv.org/abs/2104.08821>
论文 “SimCSE: Simple Contrastive Learning of Sentence Embeddings”
- <https://www.diva-portal.org/smash/record.jsf?pid=diva2%3A1684806&dswid=-528>
论文 “Semantic Re-tuning with Contrastive Tension”
- <https://arxiv.org/abs/2104.06979>
论文 “TSDAE: Using Transformer-based Sequential Denoising Auto-Encoder for Unsupervised Sentence Embedding Learning”
- <https://arxiv.org/abs/2112.07577>
论文 “GPL: Generative Pseudo Labeling for Unsupervised Domain Adaptation of Dense Retrieval”

11.2 节

- <https://github.com/huggingface/setfit>
Hugging Face 官方 GitHub 仓库中的 SetFit 项目
- <https://arxiv.org/abs/1902.00751>
论文 “Parameter-Efficient Transfer Learning for NLP”
- <https://adapterhub.ml/>
AdapterHub，适配器网站
- <https://arxiv.org/abs/2007.07779>
论文 “AdapterHub: A Framework for Adapting Transformers”
- <https://arxiv.org/abs/2303.16199>
论文 “LLaMA-Adapter: Efficient Fine-tuning of Language Models with Zero-init Attention”
- <https://arxiv.org/abs/2012.13255>
论文 “Intrinsic Dimensionality Explains the Effectiveness of Language Model Fine-Tuning”
- <https://arxiv.org/abs/2305.14314>
论文 “QLoRA: Efficient Finetuning of Quantized LLMs”

12.3 节

- <https://huggingface.co/TinyLlama/TinyLlama-1.1B-intermediate-step-1431k-3T>
Hugging Face 模型库中 TinyLlama-1.1B 模型的页面
- <https://huggingface.co/TinyLlama/TinyLlama-1.1B-Chat-v1.0>
Hugging Face 模型库中 TinyLlama-1.1B-Chat-v1.0 模型的页面
- https://huggingface.co/datasets/HuggingFaceH4/ultrachat_200k
Hugging FaceH4/ultrachat_200k 数据集的介绍页面
- <https://github.com/bitsandbytes-foundation/bitsandbytes>
bitsandbytes 包的 GitHub 仓库
- <https://github.com/huggingface/peft>
peft 库的 GitHub 仓库
- <https://magazine.sebastianraschka.com/p/practical-tips-for-finetuning-llms>
用 LoRA 微调 LLM 的更多相关内容，作者分享了通过大量实验得出的实用技巧，并解答了关于 LoRA 的常见问题

12.4 节

- <https://github.com/hendrycks/test>
MMLU 基准测试的资源
- <https://github.com/sylinrl/TruthfulQA>
TruthfulQA 基准测试的资源
- <https://huggingface.co/datasets/gsm8k>
GSM8k 基准测试的资源
- <https://rowanzellers.com/hellaswag/>
HellaSwag 基准测试的资源
- <https://github.com/openai/human-eval>
HumanEval 基准测试的资源
- https://huggingface.co/spaces/HuggingFaceH4/open_llm_leaderboard
关于 LLM 的排行榜

12.7 节

- <https://huggingface.co/datasets/argilla/distilabel-intel-orca-dpo-pairs>
Hugging Face 数据集库中 argilla/distilabel-intel-orca-dpo-pairs 数据集的页面

- <https://arxiv.org/abs/2403.07691>
论文 “ORPO: Monolithic Preference Optimization without Reference Model”

附录 A 原文中的其他资源推荐

- <https://newsletter.maartengrootendorst.com/p/a-visual-guide-to-reasoning-llms>
Maarten Grootendorst 的文章 “A Visual Guide to Reasoning LLMs”
- <https://www.interconnects.ai/p/deepseek-r1-recipe-for-o1>
Nathan Lambert 的文章 “DeepSeek R1’s Recipe to Replicate o1 and the Future of Reasoning LMs”
- <https://newsletter.maartengrootendorst.com/p/a-visual-guide-to-mixture-of-experts>
Maarten Grootendorst 的文章 “A Visual Guide to Mixture of Experts (MoE)”
- <https://www.youtube.com/watch?v=6PEJ96k1kiw>
Sasha Rush 的 YouTube 视频 “Speculations on Test-Time Scaling (o1)”
- https://www.youtube.com/watch?v=bAWV_yr4x4w
Yannis Kilcher 的 YouTube 视频 “[GRPO Explained] DeepSeekMath: Pushing the Limits of Mathematical Reasoning in Open Language Models”
- <https://github.com/huggingface/open-r1>
复现 DeepSeek-R1 的项目 Open R1 的 GitHub 仓库
- https://huggingface.co/blog/putting_rl_back_in_rlhf_with_rloo
文章 “Putting RL back in RLHF”
- <https://arxiv.org/abs/2211.09085>
论文 “Galactica: A Large Language Model for Science”

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